

Green Cart Ltd. – Q2 Business Performance Story

Introduction

As Green Cart Ltd. continues its rapid growth within the UK e-commerce market, understanding customer behaviour, operational efficiency, and regional sales performance has become increasingly important ahead of the Q2 business review.

To support this objective, the Data & Insight team conducted an in-depth analysis using sales, product, and customer datasets. The goal of this project was not only to measure revenue performance across regions and product categories, but also to uncover operational challenges affecting customer experience particularly delivery delays and identify opportunities for business growth.

Through data visualisation, feature engineering, and trend analysis, this report reveals how customer loyalty, discounts, product categories, and fulfilment performance influence overall business success.

Preparing the Data for Analysis

Before any meaningful analysis could begin, the datasets required careful cleaning and standardisation to ensure accuracy and reliability.

Three CSV datasets: sales, products, and customers, were imported and examined. During this process, several data quality issues were identified, including missing values, inconsistent text labels, and duplicate records.

Missing categorical values were replaced with “Unknown,” while missing discount values were set to 0.0 to preserve analytical consistency. Date-related fields such as signup dates and launch dates were converted into proper datetime formats, enabling accurate time-based analysis.

The team also identified duplicate order records, specifically order IDs 0515400 and 0916245, which were removed to prevent distortion in revenue calculations.

Inconsistent customer and transaction labels were then standardised. For example:

- “gld” became “Gold”
- “silver” became “Silver”
- “femle” became “Female”
- “credit card” became “Credit Card”
- “DELAYED” became “Delayed”

```
# Confirm Merged Successfully
```

```
print(final_merged_data.head())
```

	order_id	customer_id	product_id	quantity	unit_price	order_date	\
0	0966977	C00397	P0022	3	39.25	2025-06-07	
1	0696648	C00236	P0023	5	18.92	2025-06-07	
2	0202644	C00492	P0011	1	29.68	2025-06-07	
3	0501803	C00031	P0003	1	32.76	2025-06-07	
4	0322242	C00495	P0016	1	47.62	2025-06-07	

	delivery_status	payment_method	region_x	discount_applied	\
0	Delivered	File display PayPal	Central	0.00	
1	Delayed	Credit Card	North	0.00	
2	Delivered	Bank Transfer	North	0.15	
3	Cancelled	Credit Card	Central	0.20	
4	Delayed	Credit Card	West	0.20	

	product_name	category	launch_date	base_price	supplier_code	\
0	Cleaning Product 86	Cleaning	2024-04-10	19.77	S924	
1	Outdoors Product 32	Outdoors	2025-01-18	36.04	S907	
2	Kitchen Product 53	Kitchen	2025-03-25	11.08	S645	
3	Cleaning Product 85	Cleaning	2024-07-15	7.92	S745	
4	Cleaning Product 53	Cleaning	2025-02-20	26.71	S715	

	email	signup_date	gender	region_y	loyalty_tier
0	jacquelinemartin@mills-logan.com	2024-08-09	Female	North	Silver
1	georgeweber@morgan.com	2024-06-08	Other	North	Gold
2	lcook@walters-smith.com	2024-09-25	Male	Central	Gold
3	smoore@gmail.com	2024-07-28	Female	Central	Gold
4	mcclainkatherine@hotmail.com	2024-09-18	Male	Central	Gold

Although these corrections may appear minor, they were essential for ensuring reliable joins, accurate visualisations, and trustworthy insights.

Creating New Business Insights Through Feature Engineering

To gain deeper insight into customer and operational behaviour, several new analytical features were created.

- **Revenue:** A revenue metric was calculated to measure total transaction income, enabling clearer visibility into sales performance across categories and regions.
- **Order Week:** Orders were grouped into calendar weeks to identify weekly sales trends and support future demand forecasting.
- **Price Bands:** Products were categorised into Low, Medium, and High price bands, allowing the business to better understand customer purchasing preferences across pricing tiers.
- **Days to Order:** This metric measured the time between customer sign-up and first purchase, helping evaluate customer activation speed.
- **Email Domain:** Customer email domains were extracted to distinguish between corporate and personal buyers and support future marketing strategies.
- **Is Late:** A Boolean indicator was introduced to identify delayed deliveries, making it easier to assess fulfilment performance and customer service risks.

Together, these engineered features transformed raw transactional data into actionable business intelligence.

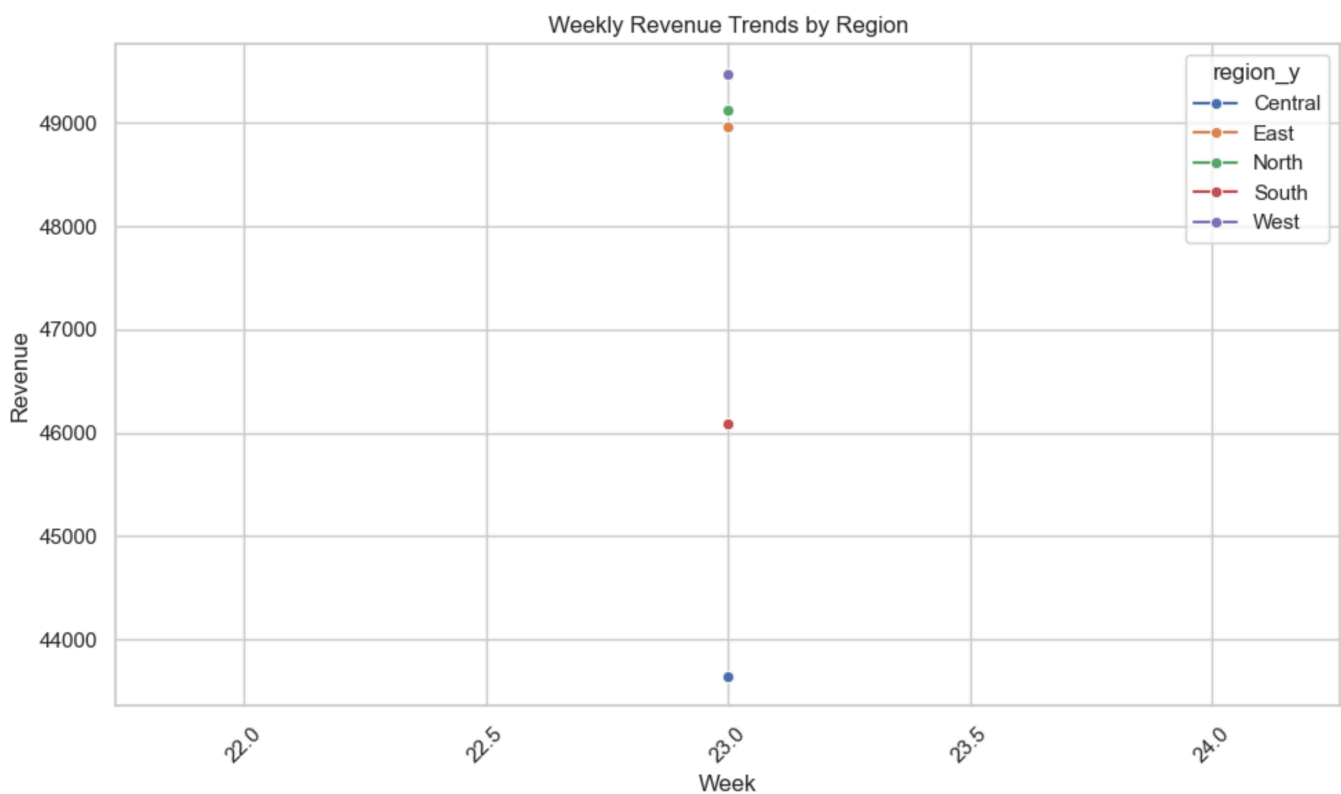
Key Business Findings

Revenue Performance Across Regions

The analysis revealed relatively stable weekly revenue patterns across all regions, with only moderate fluctuations during Week 23.

However, regional performance differences still emerged clearly:

- The East region generated the highest overall revenue.
- The Central region recorded the lowest revenue performance.

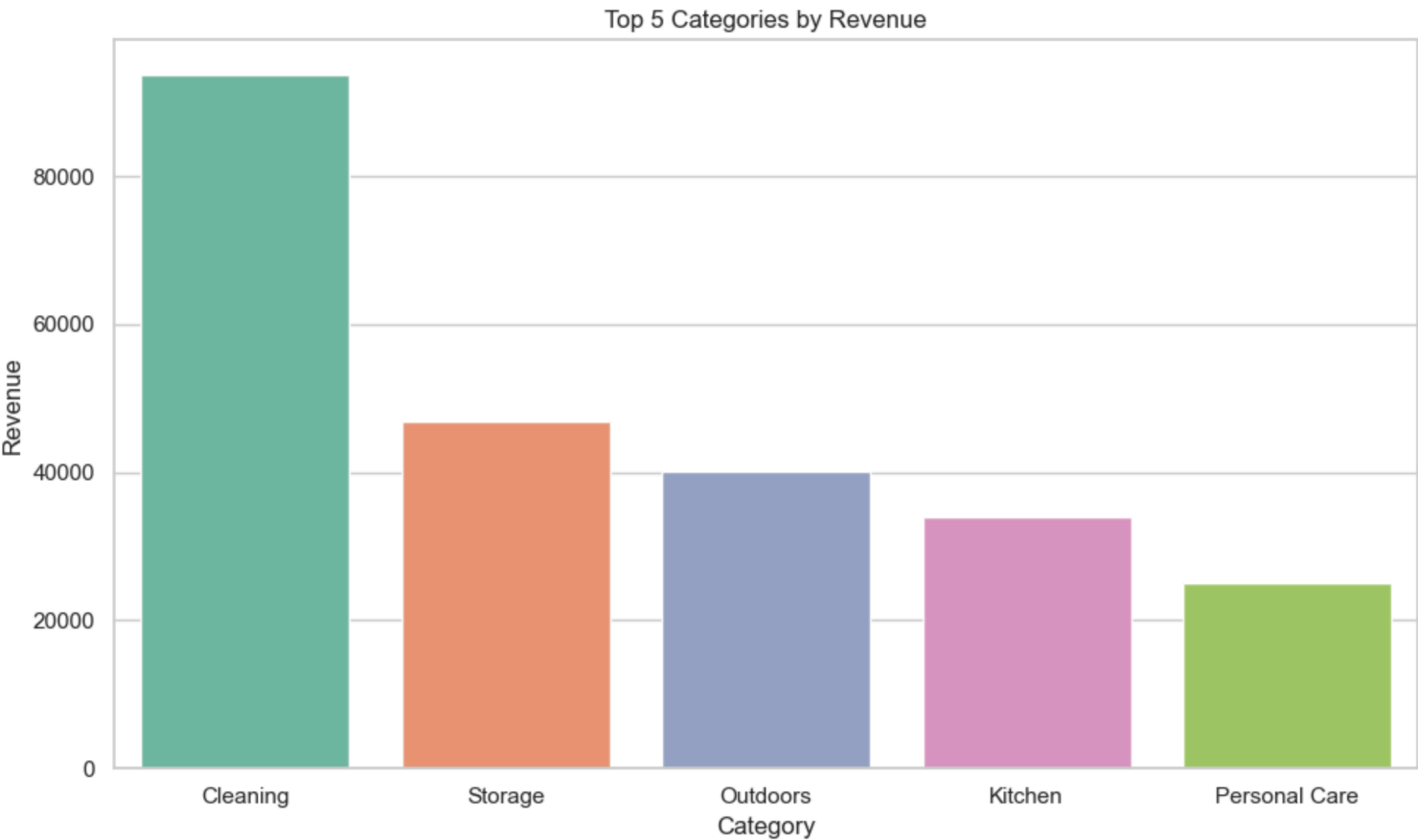


This indicates that while Green Cart maintains broad market consistency, some regions present stronger growth opportunities than others.

Product Categories Driving Revenue

One of the clearest findings was the dominance of the Cleaning category.

The Cleaning category generated revenues exceeding 80,000, significantly outperforming all other product categories. In contrast, the Personal Care category produced the lowest revenue, averaging only 20,000 overall.

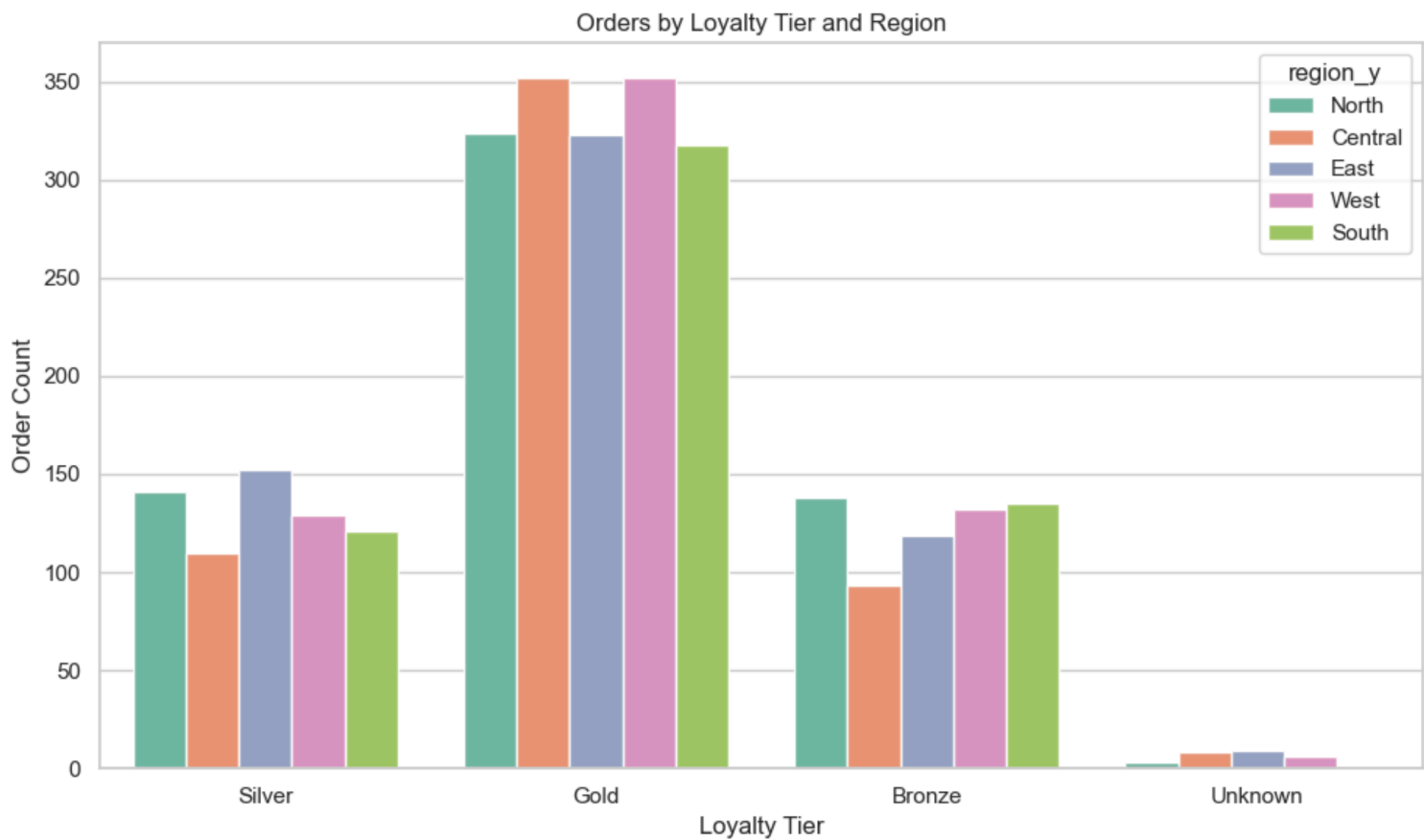


This gap highlights a major imbalance in category performance and suggests untapped potential within Personal Care products.

Customer Loyalty Value

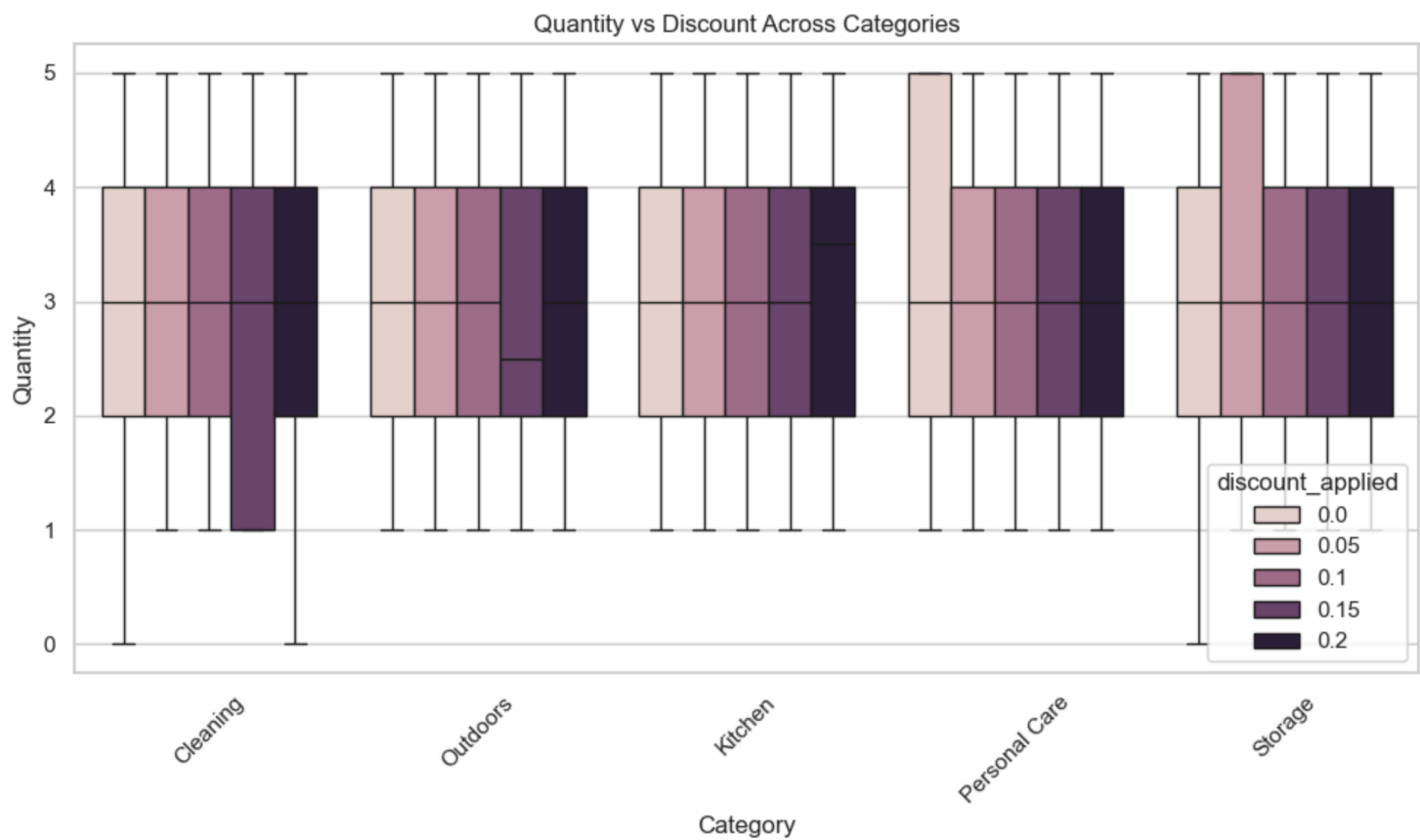
Customer loyalty analysis showed that Gold-tier customers contribute the highest overall value to the business.

This finding confirms the importance of customer retention strategies and demonstrates that loyal customers remain a major driver of long-term profitability.



Discounts Have Limited Influence on Sales Volume

The relationship between discounts and quantity sold was weaker than expected.



Across most categories, sales volume remained relatively stable regardless of discount levels. This suggests that aggressive discounting may not significantly increase purchasing behaviour and could unnecessarily reduce profit margins.

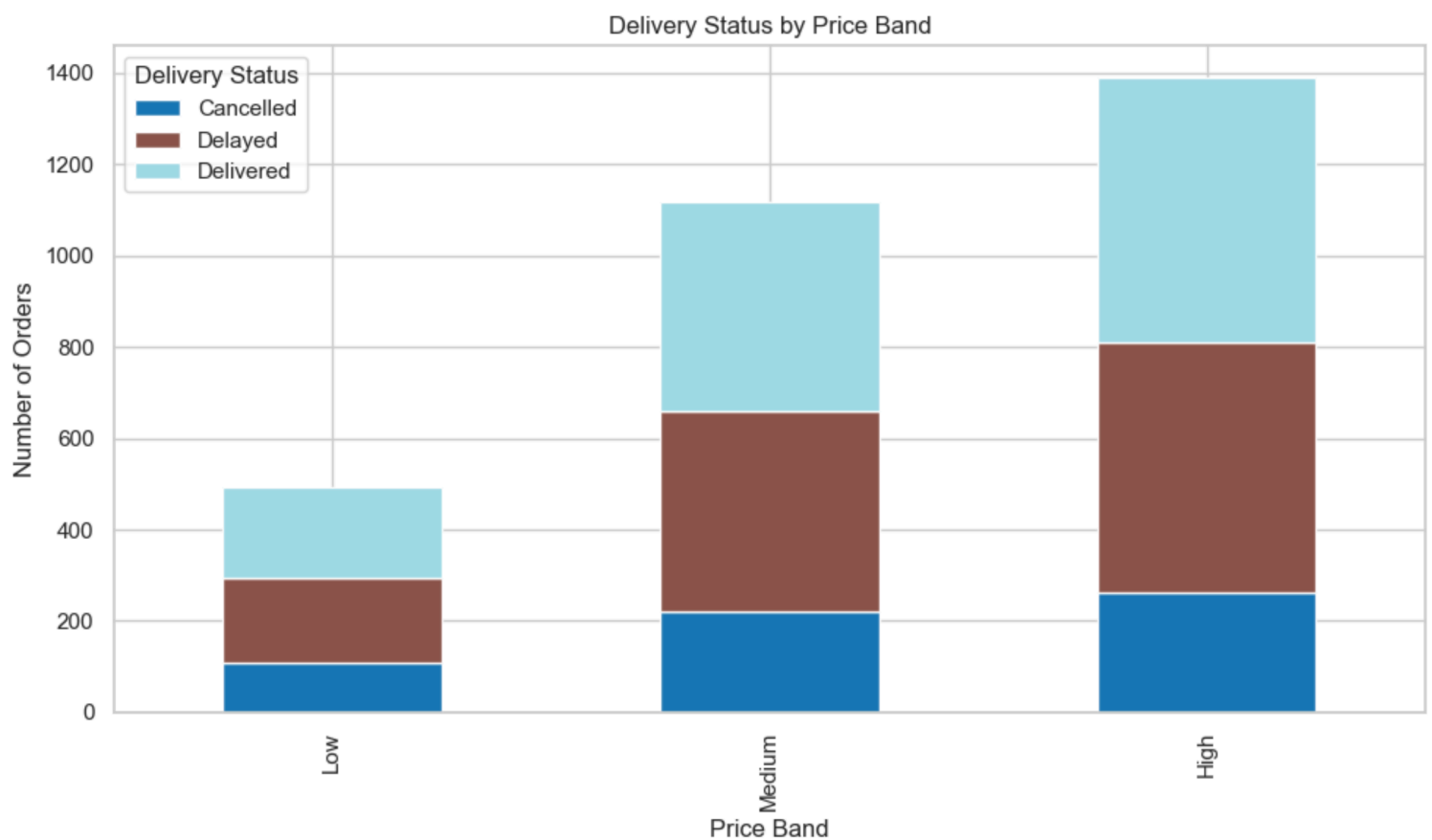
Delivery Delays Are a Major Operational Concern

Perhaps the most critical operational finding relates to delivery performance.

Several regions reported delay rates exceeding 40%:

- West region
- Central region
- South region

The North region followed closely behind at 39%, while the East region performed best with a delay rate of 35%.



Although the East region achieved the strongest revenue performance, the widespread delay issues across other regions represent a significant risk to customer satisfaction and long-term loyalty.

Business Impact and Interpretation

The findings suggest that Green Cart's strongest challenge is not generating demand it is maintaining operational efficiency while scaling rapidly.

The company already has strong-performing product categories and valuable loyal customers. However, delivery reliability threatens customer experience, particularly in high-performing regions such as the West.

	category	region_y	revenue
1	Cleaning	East	20533.6275
4	Cleaning	West	19821.9695
2	Cleaning	North	19693.1075
3	Cleaning	South	17081.0815
0	Cleaning	Central	15739.2890
24	Storage	West	10201.7605
20	Storage	Central	9797.5680
23	Storage	South	9530.4005
21	Storage	East	9017.1015
14	Outdoors	West	8990.5160
12	Outdoors	North	8923.7850
22	Storage	North	7920.0025
10	Outdoors	Central	7470.6555
11	Outdoors	East	7448.8280
7	Kitchen	North	7385.8370
6	Kitchen	East	7210.1390
8	Kitchen	South	7164.7470
5	Kitchen	Central	6639.2535
13	Outdoors	South	6569.0740
18	Personal Care	South	5593.9120
9	Kitchen	West	5471.7590
16	Personal Care	East	5053.9605
19	Personal Care	West	4987.2335
17	Personal Care	North	4851.9200
15	Personal Care	Central	3941.0195

At the same time, the underperformance of the Personal Care category presents a valuable growth opportunity. Since discounting shows limited influence on purchasing behaviour, future growth may depend more on targeted marketing and product positioning than price reductions.

Strategic Recommendations

1. Expand Personal Care Sales in High-Performing Regions

The Personal Care category currently generates the lowest revenue despite strong purchasing activity in regions such as the East and West.

Green Cart should introduce targeted promotional campaigns focused on Personal Care products in these regions through:

- Bundle offers with Cleaning and Kitchen products
- Personalised digital marketing
- Loyalty-based promotions for Gold-tier customers

This approach could unlock significant untapped revenue potential.

2. Improve Delivery Reliability

Delivery delays represent the company's most urgent operational issue.

To address this challenge, Green Cart should consider:

- Establishing local distribution centres in high-delay regions
- Improving logistics coordination
- Introducing proactive delivery notifications and delay communication systems

Improving transparency and delivery speed would likely increase customer trust and retention.

3. Strengthen Data Quality Controls

The analysis also exposed several data quality risks, including missing values and inconsistent entries across customer and sales records.

To improve future reporting accuracy, Green Cart should:

- Make critical fields mandatory during form submissions
- Introduce dropdown validation for loyalty tiers and payment methods
- Implement automated data quality monitoring processes

Better data quality will lead to more reliable analytics and stronger business decisions.

Conclusion

This analysis demonstrates that Green Cart Ltd. is positioned strongly for continued growth, supported by stable regional revenue performance, highly valuable loyal customers, and a dominant Cleaning product category.

However, operational inefficiencies particularly delivery delays pose a growing risk to customer satisfaction. Simultaneously, underperforming categories such as Personal Care offer significant expansion opportunities if supported by targeted marketing strategies.

By improving fulfilment reliability, strengthening customer engagement, and maintaining high-quality data practices, Green Cart can enhance both operational performance and long-term profitability in future quarters.